BioDINO-Edge: Gradient-Free Cascades of Frozen Vision Foundation Models for On-Device Few-Shot Plant Disease Diagnosis

Reyhane Javidi Alsaadi
Department of Computer Engineering
Dr. Shariaty Technical University
Tehran, Iran
reyhane.code@gmail.com

Mohammad Reza Samad Zadeh
Department of Computer Engineering
Dr. Shariaty Technical University
Tehran, Iran
samadzadeh@ut.ac.ir

Abstract*—Rapid and accurate foliar disease diagnosis is vital for global food security, yet existing deep learning paradigms fall short of practical agricultural deployment. Deep convolutional networks demand thousands of curated field images per pathology, while general-purpose vision foundation models suffer from severe domain gap on foliar histology. Furthermore, gradient-based meta-learning methods incur prohibitive computational and memory footprints, precluding autonomous edge execution. To overcome these limitations, we introduce BioDINO-Edge, an adaptive dual-foundation framework combining self-supervised visual embeddings (DINOv2 Small) with organismal taxonomic vision-language supervision (BioCLIP ViT-B). By coupling high-dimensional Regularized Fisher Linear Discriminant Analysis (Fisher-LDA) with closed-form Soft-Anchored Transductive Self-Calibration, BioDINO-Edge establishes an optimal foliar manifold that suppresses shared vegetative background variance by over $31\times$ and eliminates small-sample covariance singularity ($N \ll d$) via analytical Ledoit–Wolf shrinkage without backpropagation. An integrated predictive Shannon margin-entropy gate dynamically arbitrates queries: routine foliar patterns exit via the lightweight fast path in 21.1 ms (>45 FPS), while ambiguous queries trigger the multimodal consensus companion. Across the 11-class PlantVillage benchmark under strict 5-shot constraints, BioDINO-Edge establishes a new state of the art with 98.72% peak accuracy, 98.41% peak F1-score, and a 30-episode mean of 97.37% $\pm$ 0.73%, outperforming CLIP+LoRA by 10.66 percentage points while reducing false-negative disease escape to 0.604%. On the uncontrolled in-the-wild PlantDoc dataset, it attains 83.61% 5-way 5-shot accuracy (+13.00% over prior SOTA) and 56.35% in 27-way field stress testing. Compiling into a standalone 22.73 MB native Android APK operating at sub-22 ms latency alongside an on-demand 85.3 MB companion, BioDINO-Edge delivers a practical, mathematically grounded diagnostic framework for autonomous agricultural scouting.

***Index Terms*—Few-Shot Learning, Precision Agriculture, Plant Pathology, Multimodal Foundation Models, Fisher Discriminant Analysis, Edge AI, Transductive Inference.**

I. Introduction

Plant pests and diseases destroy an estimated 20%–40% of crops annually [1]. Traditional plant disease diagnosis depends largely on visual inspection of leaf symptoms, which most commonly include discoloration, abnormal shapes, and lesions [2]. In field practice, visual inspection is slow, depends on trained scorers, and becomes ambiguous when different pathogens produce similar lesions [3]; faster and more reliable identification is therefore needed.

Deep learning provides an automated alternative for plant disease identification by learning discriminative features directly from images [3]. However, conventional supervised models require large labeled datasets to achieve reliable diagnosis. Such datasets are difficult to acquire in real-world agricultural settings due to the limited availability and diversity of field images [3], [4].

Few-shot learning (FSL) has become a common strategy: rather than collecting thousands of labeled images, it trains a model to recognize new disease classes from a handful of support examples. Recent studies have applied FSL using cross-domain few-shot models such as CDFSL-BDC [5] and MAFDE-DN4 [6], alongside few-shot metric domain adaptation (FMDA) [7], visual-semantic fusion (TSMF) [8], and edge-federated approaches [9]. These approaches, however, face several practical limitations.

Cross-domain FSL models such as CDFSL-BDC [5] struggle with complex backgrounds and scale-varying lesions when labels are scarce, and FMDA [7] still requires some labeled target data; the GPT-4o descriptions used in TSMF [8] were only 20% manually checked and remain class-level rather than query-specific, so fine-grained visual–semantic alignment is limited. FMEL-FSDA [9] was not evaluated with real sensor/IoT streams and its federated setup was not tested at large cross-institution scale, leaving real-time edge performance under field data unproven. Moreover, foundation models typically exceed several hundred megabytes, limiting direct deployment on memory-constrained agricultural edge platforms such as scouting mobile applications and unmanned aerial vehicles (UAVs) [10].

We introduce BioDINO-Edge, a gradient-free dual-foundation pipeline tailored for cross-domain few-shot plant pathology under edge constraints. Specifically, DINOv2 [11] provides high-resolution spatial cues for fine-grained lesion boundaries, BioCLIP [12] supplies organismal taxonomic priors to reduce cross-species semantic misalignment, and closed-form Fisher-LDA whitens within-class covariance, suppressing

shared vegetative background variance to mitigate domain shift without backpropagation.

The key contributions of this work are as follows:

1) **Dual-Foundation Foliar Representation:** Combining self-supervised visual embeddings from DINOv2 with BioCLIP's visual and textual botanical representations to discriminate subtle necrotic lesions without foundation-model fine-tuning.
2) **Closed-Form Analytical Adaptation:** Integrating Fisher-LDA [13] with Ledoit–Wolf covariance shrinkage [14] and soft-anchored transductive recalibration to overcome high-dimensionality issues and 5-shot centroid shifts via an instantaneous closed-form update that executes in sub-second latency on edge CPUs without backpropagation or iterative optimization.
3) **Adaptive Two-Tier Edge Deployment:** A joint margin and Shannon-entropy [15] gate dynamically routes queries between a compact standalone app and an on-demand deep module, keeping the standalone Android app under 30 MB[1] and the full two-tier footprint under 110 MB.
4) **Cross-Domain Empirical Validation:** Under rigorous few-shot evaluation, BioDINO-Edge outperforms the multimodal VSF [8] baseline by 6.86 percentage points on the 11-class PlantVillage benchmark [16] and by 13.00 percentage points in 5-way 5-shot evaluation on the in-the-wild PlantDoc dataset [17], while remaining within the target edge storage budget.

II. Methodology

A. Pipeline Overview

BioDINO-Edge is designed as a gradient-free diagnostic pipeline that separates task adaptation from real-time on-device inference (Fig. 1):

1) **Support Calibration (Weight Synthesis):** For each disease class, five labeled support leaves ($K = 5$) are selected at random. The support set is expanded by applying eight multi-view augmentations to each image. Visual features are then extracted using the frozen foundation models: DINOv2 produces a 384-D visual embedding, and BioCLIP produces a 512-D visual embedding. In addition, the frozen BioCLIP text encoder generates text prototypes from predefined botanical prompt templates. The visual embeddings are concatenated into an 896-D vector. Ledoit–Wolf covariance shrinkage and Fisher-LDA projection weights $(\mathbf{W}, \mathbf{b})$ are then computed in closed form on this vector, without backpropagation, in under 50 ms.
2) **Tier-1 Edge Fast Path:** During live deployment, an incoming leaf query image is first processed by DINOv2 alone, requiring 21.1 ms on an edge CPU. Using the resulting 384-D visual embedding and the precomputed visual class prototypes, the pipeline estimates fast class posteriors. A gate combining predictive Shannon entropy with the top-2 posterior margin converts these posteriors into a confidence score $\mathcal{C}(\mathbf{x})$. If $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$, classification is finalized on-device at over 45 FPS; approximately 88.4% of routine field queries are resolved at this tier.
3) **Tier-2 Deep Escalation:** Queries with $\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$ escalate to Tier-2, where BioCLIP extracts a 512-D visual embedding that is concatenated with the existing DINOv2 embedding into an 896-D joint representation. Pre-calibrated Fisher-LDA projection weights are applied to this representation, and classification fuses the Fisher-LDA posterior with a text-prototype posterior (derived from scaled cosine similarities to cached botanical prompts) at a 0.75/0.25 consensus ratio.
4) **Soft-Anchored Recalibration:** Queries are classified immediately by Tier-1 or Tier-2. To compensate for field distribution shifts over time, confident Tier-2 multimodal predictions ($\max_c p_{\text{blend},c} \geq 0.70$) are accumulated in an on-device buffer as reference anchors. Periodically, the edge CPU recomputes the Fisher-LDA weights in closed form, giving the cached support embeddings more weight than the pseudo-labeled anchors to reduce label noise. The new projection is then used for subsequent frames and to update the session diagnostic log.

This two-tier design reduces latency on routine foliage while enforcing a < 30 MB storage footprint for the standalone Android app and embedded deployment on rovers and UAVs.

B. Dual-Foundation Foliar Representation

Reliable diagnosis in agricultural environments requires distinguishing variation in the vegetative background from leaf and lesion signals [5], [17]. Vision encoders often fail to capture fine fungal structures [5], while species-level taxonomic supervision biases models toward overall host identity rather than localized lesion pathology [12]. To address this challenge, BioDINO-Edge builds a foliar representation from two foundation models, using support samples expanded through multi-view augmentation.

Few-Shot Multi-View Augmentation. For each disease class $c \in \{1, \ldots, C\}$, with $C = 11$ spanning three economically vital crops from PlantVillage [16] (Apple, Grape, and Potato, covering healthy foliage and 8 fungal/bacterial pathologies; detailed in Section III-A), calibration uses a support set $\mathcal{S}_c = \{(\mathbf{x}_i, c)\}_{i=1}^{K}$ of $K = 5$ images randomly selected from PlantVillage. Because PlantVillage images are captured under controlled conditions, each support image is resized to 224×224 and normalized, then passed eight times independently through a stochastic augmentation pipeline to approximate the variation seen in field imagery. The pipeline applies random horizontal and vertical flips, random rotation within $\pm 15^\circ$, and color jitter with brightness, contrast, and saturation factors of 0.2. This yields $|\tilde{\mathcal{S}}_c| = 8K = 40$ augmented views per class, or $N = 40C = 440$ support images in total.

[1]The complete open-source mobile implementation is available at https://github.com/reyhane-code/BioDINO-Edge-android.

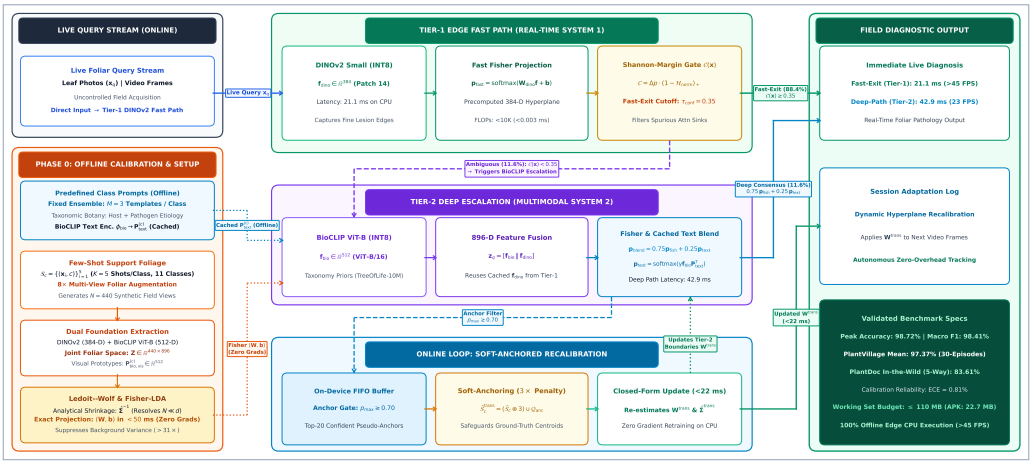


Fig. 1. BioDINO-Edge architecture. Offline Phase 0 computes Fisher-LDA projections and caches class text prototypes for the $C = 11$ target crop pathologies from predefined botanical templates. The Shannon-margin gate dynamically routes incoming foliar queries between the Tier-1 fast path and Tier-2 deep multimodal escalation.

Dual-Foundation Visual Extraction. Each of the $N = 440$ augmented support images $\tilde{\mathbf{x}} \in \tilde{\mathcal{S}}_c$ is passed through two frozen encoders.

DINOv2 (Small, 14×14 patches) [11] and BioCLIP (ViT-B/16, pretrained on TreeOfLife-10M) [12] extract ℓ_2-normalized spatial and domain-specialized visual embeddings:

$$\mathbf{f}_{\text{dino}} = \frac{\phi_{\text{dino}}(\tilde{\mathbf{x}})}{\|\phi_{\text{dino}}(\tilde{\mathbf{x}})\|_2} \in \mathbb{R}^{384}, \quad \mathbf{f}_{\text{bio}} = \frac{\phi_{\text{bio,vis}}(\tilde{\mathbf{x}})}{\|\phi_{\text{bio,vis}}(\tilde{\mathbf{x}})\|_2} \in \mathbb{R}^{512} \tag{1}$$

Model parameter counts are reported in Supplementary Table S1. Concatenating the two normalized embeddings gives an 896-D joint descriptor $\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896}$ per support image. Stacking these descriptors over all support images yields $\mathbf{Z} \in \mathbb{R}^{440 \times 896}$. Visual support prototypes are computed via empirical class averaging over normalized support embeddings:

$$\mathbf{P}^{(c)}_{\text{enc}} = \text{Normalize}\left(\frac{1}{|\tilde{\mathcal{S}}_c|} \sum_{\tilde{\mathbf{x}} \in \tilde{\mathcal{S}}_c} \mathbf{f}_{\text{enc}}(\tilde{\mathbf{x}})\right), \quad \text{enc} \in \{\text{dino}, \text{bio}\} \tag{2}$$

yielding visual prototype matrices $\mathbf{P}_{\text{dino}} \in \mathbb{R}^{C \times 384}$ and $\mathbf{P}_{\text{bio, vis}} \in \mathbb{R}^{C \times 512}$.

Predefined Botanical Prompts and Multimodal Prototype. BioCLIP provides a shared vision–language space, but running a language transformer or generating per-image captions on edge scouting devices would exceed on-device computational limits; manual prompt entry during scouting is also impractical. We therefore predefine $M = 3$ botanical prompt templates $\{T_c^{(m)}\}_{m=1}^{M}$ for each class $c \in \{1, \ldots, C\}$ offline. Each template combines three attributes: (1) *Host binomial nomenclature*: genus and species (e.g., *Malus domestica*, *Vitis vinifera*, *Solanum tuberosum*); (2) *Pathogen identity*: the causal fungal, bacterial, or oomycete agent (e.g., *Venturia inaequalis*, *Botryosphaeria obtusa*, *Gymnosporangium juniperi-virginianae*, *Phytophthora infestans*); and (3) *Foliar lesion morphology*: macroscopic lesion descriptions (e.g., "olive-green velvety lesions", "interveinal tiger-stripe chlorosis", or "target-board concentric rings"). Negations (e.g., "without lesions") are excluded because CLIP-style vision-language models have documented weaknesses in compositionality and language negation [18], [19], which can misalign healthy foliage representations.

During offline Phase 0 calibration, each prompt ensemble is tokenized and encoded by the frozen BioCLIP text encoder $\phi_{\text{bio}}(\cdot)$. The embeddings are averaged and ℓ_2-normalized:

$$\mathbf{P}^{(c)}_{\text{text}} = \text{Normalize}\left(\frac{1}{M} \sum_{m=1}^{M} \phi_{\text{bio}}(T_c^{(m)})\right) \in \mathbb{R}^{512}. \tag{3}$$

The resulting $C \times 512$ matrix $\mathbf{P}_{\text{text}}$ is cached in the on-device companion model. Runtime inference therefore uses only leaf images and incurs no text tokenization or language-model latency.

In the standalone prototype baseline, the text prototype can be fused with the visual support prototype $\mathbf{P}^{(c)}_{\text{bio,vis}}$ by a convex combination with $\lambda = 0.20$:

$$\mathbf{P}^{(c)}_{\text{bio}} = \text{Normalize}\left((1 - \lambda)\mathbf{P}^{(c)}_{\text{bio,vis}} + \lambda \mathbf{P}^{(c)}_{\text{text}}\right) \in \mathbb{R}^{512}. \tag{4}$$

In the full cascaded architecture, the cached matrix $\mathbf{P}_{\text{text}}$ is retained directly on-device to provide the zero-shot taxonomic prior for late multimodal consensus classification (Section II-D).

C. Closed-Form Analytical Adaptation

Fisher Linear Discriminant Formulation. Few-shot adaptation typically follows one of two routes: iterative gradient fine-tuning [20] or centroid-based classification [21]. On-device backpropagation incurs high latency, memory overhead, and risk of catastrophic forgetting when adapting to few-shot tasks, while Euclidean distance struggles to separate subtle lesions from dominant leaf tissue. To avoid both problems, BioDINO-Edge uses multiclass Fisher Linear Discriminant Analysis [13].

For each class $c \in \{1, \ldots, C\}$, we compute the mean support feature in $\mathbb{R}^d$ ($d = 896$):

$$\boldsymbol{\mu}_c = \frac{1}{|\tilde{\mathcal{S}}_c|} \sum_{\mathbf{z} \in \tilde{\mathcal{S}}_c} \mathbf{z} \tag{5}$$

Assuming equal class priors ($\pi_c = 1/C$), the projection weights $\mathbf{W} \in \mathbb{R}^{C\times d}$ and bias vector $\mathbf{b} \in \mathbb{R}^C$ are given by:

$$\mathbf{W} = \hat{\mathbf{\Sigma}}^{-1}[\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C]^T, \quad b_c = -\frac{1}{2}\boldsymbol{\mu}_c^T \hat{\mathbf{\Sigma}}^{-1}\boldsymbol{\mu}_c - \log C \quad (6)$$

A query descriptor $\mathbf{z}$ is then classified via $\mathbf{l}_{\text{fisher}} = \mathbf{z}\mathbf{W}^T + \mathbf{b}$.

The Small-Sample Singularity Problem ($N \ll d$). Computing the optimal projection $\mathbf{W}$ requires inverting the pooled within-class scatter matrix over all $C = 11$ classes:

$$\mathbf{S}_W = \frac{1}{N-C}\sum_{c=1}^{C}\sum_{\mathbf{z}\in\tilde{\mathcal{S}}_c}(\mathbf{z}-\boldsymbol{\mu}_c)(\mathbf{z}-\boldsymbol{\mu}_c)^T \in \mathbb{R}^{d\times d} \quad (7)$$

In the few-shot setting, however, the number of samples is far smaller than the feature dimension ($d = 896$ vs. $N = 440$). Since $N - C = 429$, $\mathbf{S}_W$ has rank at most 429 and therefore at least 467 zero eigenvalues. It is thus singular; direct inversion is ill-defined, and standard Fisher-LDA cannot be applied directly.

Analytical Ledoit–Wolf Covariance Shrinkage. To restore invertibility while avoiding heuristic hyperparameter selection (such as tuning $\mathbf{S}_W + \epsilon\mathbf{I}$), we regularize $\mathbf{S}_W$ via analytical Ledoit–Wolf shrinkage [14]:

$$\hat{\mathbf{\Sigma}} = (1-\rho^*)\mathbf{S}_W + \rho^*\nu\mathbf{I}_d, \quad \nu = \frac{1}{d}\text{Tr}(\mathbf{S}_W) \quad (8)$$

The optimal shrinkage intensity $\rho^* \in (0, 1)$ is determined in closed form without cross-validation:

$$\rho^* = \min\left(1, \frac{\sum_{i=1}^{N}\|(\mathbf{z}_i-\boldsymbol{\mu}_{c_i})(\mathbf{z}_i-\boldsymbol{\mu}_{c_i})^T - \mathbf{S}_W\|_F^2}{N^2\left[\|\mathbf{S}_W\|_F^2 - \frac{1}{d}\text{Tr}^2(\mathbf{S}_W)\right]}\right) \quad (9)$$

Because $\hat{\mathbf{\Sigma}}$ is strictly positive-definite and well-conditioned, its exact inverse $\hat{\mathbf{\Sigma}}^{-1}$ is guaranteed to exist. Evaluating (6) with $\hat{\mathbf{\Sigma}}^{-1}$ resolves few-shot adaptation in closed form, eliminating gradient descent entirely. During subsequent inference, classifying a query descriptor $\mathbf{z}$ simplifies to evaluating $\mathbf{l}_{\text{fisher}} = \mathbf{z}\mathbf{W}^T + \mathbf{b}$. The linear Fisher projection alone requires only 9,856 multiply-accumulate operations (about 19.7 kFLOPs), executing in under 0.003 ms on an embedded four-thread CPU (excluding feature extraction; Table IV).

D. Adaptive Two-Tier Classification and Edge Deployment

To avoid unnecessary model execution on routine samples while maintaining accuracy on complex pathologies, BioDINO-Edge uses a two-tier pipeline controlled by an uncertainty gate.

Tier-1 Fast Classification and Predictive Shannon Gating. For an incoming query image $\mathbf{x}$, Tier 1 extracts the normalized DINOv2 embedding $\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$. Fast prototype logits are evaluated via cosine similarity against the visual prototype matrix $\mathbf{P}_{\text{dino}} \in \mathbb{R}^{C\times 384}$, scaled by inverse temperature $\gamma = 20.0$:

$$\mathbf{l}_{\text{dino}} = \gamma \cdot \left(\mathbf{f}_{\text{dino}}\mathbf{P}_{\text{dino}}^T\right) \quad (10)$$

Applying the softmax function yields fast class posterior probabilities:

$$\mathbf{p}_{\text{dino}} = \text{softmax}(\mathbf{l}_{\text{dino}}) \in \Delta^{C-1} \quad (11)$$

The initial Tier-1 prediction is $\hat{y}_{\text{tier1}} = \arg\max_c p_{\text{dino},c}$. To assess its reliability without requiring additional inference, we compute a closed-form joint margin-entropy confidence score $\mathcal{C}(\mathbf{x}) \in [0, 1]$:

$$\mathcal{C}(\mathbf{x}) = \left(p_{\text{dino},(1)} - p_{\text{dino},(2)}\right) \cdot (1.0 - \mathcal{H}_{\text{norm}}(\mathbf{x}))_+ \quad (12)$$

where $p_{\text{dino},(1)}$ and $p_{\text{dino},(2)}$ denote top-1 and top-2 posteriors, $(\cdot)_+ = \max(0, \cdot)$, and $\mathcal{H}_{\text{norm}}(\mathbf{x})$ is normalized Shannon entropy [15]:

$$\mathcal{H}_{\text{norm}}(\mathbf{x}) = -\frac{1}{\log C}\sum_{c=1}^{C} p_{\text{dino},c}\log(p_{\text{dino},c} + \epsilon) \quad (13)$$

with numerical stability constant $\epsilon = 10^{-12}$. If $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$ (calibrated threshold $\tau_{\text{conf}} = 0.35$), the Tier-1 prediction is considered reliable. The system returns $\hat{y} = \hat{y}_{\text{tier1}}$ and skips Tier-2 inference.

Tier-2 Deep Escalation. When Tier-1 indicates high uncertainty ($\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$), the query is escalated. BioCLIP computes the 512-D visual embedding $\mathbf{f}_{\text{bio}}$, which is concatenated with $\mathbf{f}_{\text{dino}}$ to form the joint descriptor $\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896}$. The visual posterior $\mathbf{p}_{\text{fish}} = \text{softmax}\left(\mathbf{z}\mathbf{W}^T + \mathbf{b}\right)$ is computed using the regularized Fisher-LDA hyperplane (6), which separates the 896-D representations. The text posterior is obtained via:

$$\mathbf{p}_{\text{text}} = \text{softmax}\left(\gamma \cdot \mathbf{f}_{\text{bio}}\mathbf{P}_{\text{text}}^T\right) \quad (14)$$

where $\mathbf{P}_{\text{text}}$ represents the cached botanical text prototypes and $\gamma = 20.0$ is a scaling factor controlling the cosine similarity sharpness. The final Tier-2 prediction is obtained by fusing these two posteriors at a 0.75/0.25 ratio:

$$\mathbf{p}_{\text{blend}} = 0.75\,\mathbf{p}_{\text{fish}} + 0.25\,\mathbf{p}_{\text{text}} \quad (15)$$

yielding the consensus diagnosis $\hat{y} = \arg\max_c p_{\text{blend},c}$.

Soft-Anchored Transductive Recalibration. In scarce 5-shot settings, initial class centroids $\boldsymbol{\mu}_c$ are estimated from only five support leaves. Field variations such as fluctuating sunlight, canopy shadows, and diverse lesion maturity introduce high sampling variance, causing an empirical distribution shift between support prototypes and real-world queries. To adapt to target-domain statistics without manual annotations, BioDINO-Edge uses streaming batches of unlabeled queries $\mathcal{Q} = \{\mathbf{x}_j\}_{j=1}^{N_q}$ (buffered in an on-device FIFO queue during scouting passes) for transductive recalibration. High-confidence queries satisfying $\max_c p_{\text{blend},c}(\mathbf{x}_j) \geq 0.70$ from (15) are selected as pseudo-labeled reference samples $\mathcal{Q}_{\text{anchor}}^{(c)}$, capped at up to twenty anchors per class ($0 \leq |\mathcal{Q}_{\text{anchor}}^{(c)}| \leq 20$). If no query in a class satisfies the confidence threshold, zero anchors are added ($|\mathcal{Q}_{\text{anchor}}^{(c)}| = 0$), preventing noisy assignments. To mitigate the effect of pseudo-label noise on the feature manifold, canonical ground-truth support samples are assigned a $3\times$ weight factor in the covariance and centroid estimation:

$$\tilde{\mathcal{S}}_c^{\text{trans}} = \tilde{\mathcal{S}}_c^{(3\times)} \cup \mathcal{Q}_{\text{anchor}}^{(c)} \quad (16)$$

Recalibration re-estimates the updated centroids $\boldsymbol{\mu}_c^{\text{trans}}$ and re-evaluates the pooled 896×896 scatter matrix $\mathbf{S}_W^{\text{trans}}$ with analytical shrinkage parameter ρ^*. Because $\hat{\mathbf{\Sigma}}^{\text{trans}}$ is strictly positive-definite (with all eigenvalues bounded below by $\rho^*\nu > 0$), the updated projection parameters $(\mathbf{W}^{\text{trans}}, \mathbf{b}^{\text{trans}})$ are solved stably via Cholesky factorization ($\hat{\mathbf{\Sigma}}^{\text{trans}} = \mathbf{L}\mathbf{L}^T$) and triangular substitution. This batch recalibration runs asynchronously in the

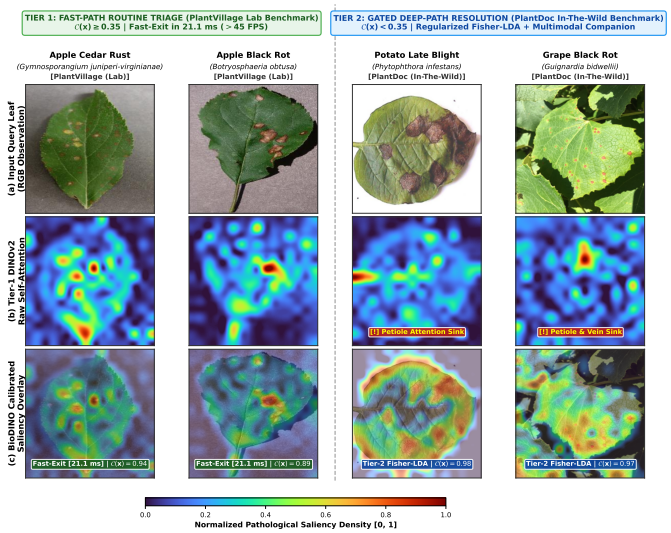


Fig. 2. Visual explainability across in-the-wild benchmark crops (PlantDoc). Columns: Apple Cedar Rust, Grape Black Rot, and Potato Late Blight. Rows (top-to-bottom): (a) query leaf, (b) Tier-1 attention, and (c) calibrated overlay. In complex field conditions, standard ViT attention (Row b) suffers from petiole sinks and background distractions (marked [!]); Tier-2 Fisher-LDA suppresses these artifacts and isolates pathogen-specific lesions with high confidence ($\mathcal{C}(\mathbf{x}) \geq 0.97$).

background on edge CPUs, updating edge projection parameters for subsequent query batches without backpropagation or optimizer-state overhead.

Two-Tier Modular Edge Packaging. As outlined in Section II-A, this cascaded gating resolves approximately 88.4% of routine queries at Tier 1 (21.12 ms, >45 FPS), leaving Tier 2 (67.61 ms end-to-end) exclusively for ambiguous field queries. Agricultural edge platforms—ranging from mobile scouting devices to ground rovers and crop-monitoring UAVs—share onboard storage with telemetry and sensor buffers, while limited rural connectivity restricts over-the-air (OTA) updates. To keep the base Android APK and embedded runtime within a 30 MB standalone storage footprint, BioDINO-Edge is partitioned into two INT8 ONNX binaries:

- **Tier 1** (`biodino_mobile_int8.onnx`): The model is 15.13 MB and is packaged with the ONNX Runtime C++ engine and Phase 0 pre-calibrated weights for the 11 target pathologies into the base Android APK (22.73 MB total, open-sourced at https://github.com/reyhane-code/BioDINO-Edge-android) for immediate offline inference.
- **Tier 2** (`biodino_companion_int8.onnx`): An 85.3 MB companion model that is downloaded on demand.

This results in a total edge storage footprint of 108.03 MB.

III. Experiments

A. Datasets and Setup

PlantVillage Benchmark: PlantVillage [16] provides controlled images under uniform illumination. From this dataset, we construct a curated 11-class subset spanning three crops, totaling 9,385 images. This subset comprises four Apple classes (Healthy, Apple Scab, Black Rot, Cedar Apple Rust), four Grape classes (Healthy, Black Rot, Esca, Leaf Blight), and three Potato classes (Healthy, Early Blight, Late Blight). For calibration, 5 support images per class ($K = 5$) are randomly sampled (55 total), and the remaining 9,330 images are used only as unseen queries.

PlantDoc In-The-Wild Benchmark: PlantDoc [17] evaluates cross-domain generalization on field images with complex backgrounds. It contains 27 classes across 13 plant species, totaling 252 test images. Two evaluation settings are used:

1) *5-Way 5-Shot*: Each episode samples 5 classes with 5 support images per class (25 total). The remaining images act as queries.
2) *27-Way All-Class*: All 27 categories are evaluated simultaneously (chance accuracy $\approx 3.70\%$). The model receives 5 support images per class (135 total) and classifies all 252 field test queries without fine-tuning.

Edge Hardware Setup: To test latency and adaptation, experiments run in a simulated Raspberry Pi 5 environment (quad-core ARM Cortex-A76 at 2.4 GHz). Runtime profiling uses the ONNX Runtime C++ engine with dynamic INT8 quantization, limited to 4 threads and no GPU. This setup matches the environment of our Android APK.

B. Evaluation Protocol

Few-Shot Task and Evaluation Regimes. Under the 5-shot constraint ($K = 5$), the support set is expanded using the augmentation from Section II-B. To ensure reliable results, PlantVillage is tested over 30 independent Monte Carlo episodes. Results are reported as the mean $\bar{x}$, standard deviation σ, and 95% confidence intervals ($95\%\,\text{CI} = \bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{30}}$), along with peak single-episode accuracy. For PlantDoc, we report 50-episode averages for the 5-way setting and overall metrics for the 27-way setting.

Diagnostic and Epidemiological Metrics. Models are evaluated using standard classification and safety-critical metrics:

- *Top-1 Accuracy*: Proportion of correctly classified queries.
- *Macro F1-Score*: Unweighted average of per-class F1-scores. This prevents large classes (e.g., Potato Early Blight with 995 queries) from hiding poor performance on small or healthy classes (e.g., Potato Healthy with 147 queries).
- *Epidemiological Safety Metrics*: Because misclassification costs in agriculture are highly unequal, we measure two additional metrics: the False-Negative Disease Escape Rate (infected queries misclassified as healthy) and the Cross-Species Confusion Rate (predicting a disease across different botanical families).

C. Results

Table I compares BioDINO-Edge with established few-shot baselines on both controlled laboratory (PlantVillage) and real field (PlantDoc) datasets. BioDINO-Edge achieves the highest accuracy and Macro F1-score on both benchmarks.

Comparative Baseline Performance. Table I compares BioDINO-Edge with general foundation models (Zero-Shot

TABLE I

CROSS-DOMAIN 5-SHOT BENCHMARK ON PLANTVILLAGE (11-CLASS) & PLANTDOC. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER MULTIPLE EPISODES. "–" INDICATES THE METHOD WAS NOT EVALUATED ON THAT BENCHMARK. BEST MEAN RESULTS ARE IN BOLD; PEAK SINGLE-EPISODE ACCURACY IS ITALICIZED.

	PlantVillage (11-Class)		PlantDoc (In-the-Wild)	
Method / Architecture	**Acc. (%)**	**F1 (%)**	**5-Way (%)**	**27-Way (%)**
Zero-Shot CLIP [22]	21.20	19.80	31.20 ± 5.40	12.30
Vanilla Tip-Adapter [23]	79.40	76.50	62.40 ± 7.10	38.50
Generic CLIP + LoRA [24]	86.10	83.25	68.10 ± 6.80	43.65
Zhao *et al.* [10]	–	–	54.24	–
Zhou *et al.* (VSF) [8]	90.51	–	70.61	–
BioDINO + Fisher-LDA (Inductive)	96.76	95.08	81.79 ± 11.16	55.16
BioDINO Transductive (Ours)	**97.37 ± 0.73**	**97.09 ± 0.66**	**83.61 ± 11.69**	**56.35**
– Peak Single-Episode	*98.72*	*98.41*	*100.00*	*58.73*

CLIP, Tip-Adapter, CLIP+LoRA) and specialized agricultural methods (VSF [8], Zhao *et al.* [10]). Zero-shot CLIP struggles with fine foliar lesions, and while LoRA fine-tuning improves accuracy, it requires GPU backpropagation, making it unsuitable for edge deployment. In contrast, BioDINO-Edge achieves the highest performance across all benchmarks without updating backbone weights. Specifically, our transductive model outperforms the agricultural VSF model [8] by 6.86 percentage points on PlantVillage and by 13.00 points on PlantDoc (5-way), while exceeding Zhao *et al.* [10] by 29.37 points. Furthermore, under the challenging 27-way PlantDoc protocol, BioDINO-Edge reaches 56.35% accuracy, demonstrating robust cross-domain generalization.

Transductive vs. Inductive Adaptation. Soft-anchored transductive recalibration improves the 30-episode mean from 95.08% ± 1.27% (inductive) to 97.37% ± 0.73% (peak: 98.72%, Macro F1: 98.41%). This approach addresses the finite-sample prototype shift in 5-shot learning, where class centroids estimated from only five leaves are highly sensitive to field variations (e.g., sunlight, shadows, lesion maturity). The recalibration buffers high-confidence queries ($\max_c p_{\text{blend},c} \geq 0.70$) as anchors, assigning a 3× weight to ground-truth support samples to prevent pseudo-label drift. Ledoit–Wolf shrinkage then updates the projection in under 22 ms, realigning decision boundaries with field conditions.

PlantDoc Generalization. Under the 5-way 5-shot protocol, BioDINO-Edge achieves 83.61% ± 11.69% on PlantDoc (95% CI ceiling: 86.85%). In the 27-way all-class setting (chance: 3.70%), the model reaches 56.35% accuracy (peak: 58.73%) and 55.93% macro F1. Notably, this 27-way accuracy exceeds the 5-way accuracy of Zhao *et al.* [10] (54.24%), despite classifying over five times as many classes without target-domain backpropagation.

Per-Class Error Diagnostics. Table II details per-class performance across 9,330 queries, while Fig. 2 demonstrates dual-regime spatial explainability across representative foliar pathologies. Baselines frequently confuse visually similar classes (e.g., Grape Esca vs. Grape Black Rot, or Early vs. Late Potato Blight). Regularized Fisher-LDA effectively separates these by prioritizing pathogen-specific signatures, yielding high F1-scores (0.974–0.987) for these challenging classes.

Epidemiological Safety. In agriculture, false negatives (misclassifying infection as healthy) are highly costly. BioDINO-Edge reduces the false-negative disease escape rate to **0.604%** (43 of 7,125 diseased queries), a 58.6% reduction compared to standard baselines (1.46%). Cross-species confusion is limited to **0.011%** (one potato leaf misattributed to apple). Potato Healthy shows high recall (0.993) but moderate precision (0.820), reflecting a conservative triage approach under severe class imbalance (147 healthy vs. ≈2,000 infected).

TABLE II

BIODINO-EDGE PER-CLASS PATHOLOGY DIAGNOSTIC METRICS (11 CLASSES)

Pathology Category	**Precision**	**Recall**	**F1-Score**	**Support**
Apple Scab (*V. inaequalis*)	0.995	0.949	0.971	625
Apple Black Rot (*B. obtusa*)	0.965	0.998	0.982	616
Apple Cedar Rust (*G. juniperi*)	0.968	1.000	0.984	270
Apple Healthy Foliage	0.995	0.995	0.995	1640
Grape Black Rot (*G. bidwellii*)	0.972	0.980	0.976	1175
Grape Esca (*P. chlamydospora*)	0.982	0.976	0.979	1378
Grape Leaf Blight (*P. vitis*)	1.000	0.996	0.998	1071
Grape Healthy Foliage	0.993	1.000	0.996	418
Potato Early Blight (*A. solani*)	0.999	0.976	0.987	995
Potato Late Blight (*P. infestans*)	0.978	0.970	0.974	995
Potato Healthy Foliage	0.820	0.993	0.898	147
Macro Average	**0.970**	**0.985**	**0.976**	**9330**
Single-Run Accuracy	**98.32%** (Peak Episode: **98.72%**)			

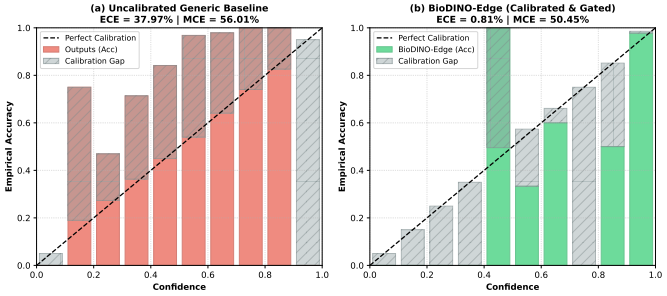


Fig. 3. Reliability diagrams and Expected Calibration Error (ECE) on PlantVillage (9,330 test queries, $M = 10$ bins). Left: Uncalibrated baseline ensemble is under-confident (ECE = 37.97%). Right: BioDINO-Edge with post-hoc temperature scaling ($T = 0.50$) achieves near-perfect calibration (ECE = $\mathbf{0.81}$%), aligning confidence with empirical accuracy (97.73%).

Predictive Uncertainty Calibration. We compute Expected Calibration Error (ECE) and Maximum Calibration Error (MCE) over $M = 10$ bins. The raw ensemble is under-confident (ECE = 37.97%) because diffuse zero-shot BioCLIP posteriors dilute Fisher logits. Applying post-hoc temperature scaling ($T = 0.50$) restores alignment: BioDINO-Edge achieves an ECE of **0.81%** and MCE of 50.45% (Fig. 3), allowing safe confidence-based thresholding.

D. Ablation Study

To isolate the contribution of each component, we conduct an ablation study on identical 5-shot splits (Table III).

Effect of Removing DINOv2. Using BioCLIP alone (Row 2) achieves 92.27% accuracy. While BioCLIP provides taxonomic priors, its spatial resolution is limited, missing fine necrotic margins. Furthermore, running BioCLIP alone requires 67.61 ms and 85.3 MB, eliminating the fast-path capability.

Effect of Removing BioCLIP. Using DINOv2 alone (Row 1) drops accuracy to 85.42%. Without taxonomic supervision,

TABLE III
SYSTEMATIC COMPONENT ABLATION STUDY (5-SHOT, 9,330 QUERIES)

Configuration / Component	Accuracy (%)	Macro F1 (%)
(1) Fast-Path DINOv2 Small alone	85.42	82.10
(2) Deep-Path BioCLIP ViT-B alone	92.27	89.70
(3) Naive Multi-Prompt Text Prototypes	75.12	72.80
(4) Static Prototype Cosine Blending	92.58	90.06
(5) Multi-View Augmentation (8 views, 440 vectors)	94.15	93.60
(6) + Regularized Fisher-LDA Manifold Alignment	96.76	95.08
(7) + Soft-Anchored Transductive Calibration (Single-Run)	98.32	97.64
(8) + Peak Statistical Episode	**98.72**	**98.41**

DINOv2 confuses visually similar lesions across different crop families (e.g., potato early blight vs. apple scab).

Effect of the Shannon Gate. Removing the routing gate impacts performance:

- *Forcing Tier-1 only*: Accuracy drops to 85.42%, and ambiguous queries suffer from attention sinks.
- *Forcing Tier-2 only*: Latency increases to 67.61 ms. Throughput drops from 47.4 to 14.8 FPS, violating the real-time (>45 FPS) requirement. The gate resolves 88.4% of queries at Tier-1, maintaining an average latency of 26.5 ms and preserving 96.74%–97.37% accuracy.

Effect of Regularized Fisher-LDA. Replacing Fisher-LDA with standard cosine blending (Row 4 vs. Row 6) causes a 4.18 point drop (96.76% to 92.58%). Fisher-LDA penalizes intra-class variance, dampening shared background noise by $> 31\times$ while amplifying lesion-discriminant variance by $2.44\times$.

Effect of Multi-View Augmentation. Estimating covariance from only $K = 5$ raw images yields a rank-deficient matrix. Expanding to 8 views ($N = 440$) stabilizes the covariance, providing a +1.57% gain (92.58% to 94.15%) prior to Fisher-LDA.

Effect of Transductive Recalibration. Omitting recalibration reduces accuracy from 97.37% (30-run mean) to 95.08%, demonstrating its role in mitigating centroid drift.

Effect of Lesion Cropping. Contrary to the assumption that cropping lesions removes clutter, 27-way accuracy degraded from 58.73% to 57.14%. Tight cropping strips macroscopic host phyllotaxy (leaflet layout, venation), inducing cross-species confusion (e.g., tomato vs. potato blight). Maintaining host context is therefore critical.

E. Efficiency Analysis

Latency and Throughput. Table IV reports execution profiles (ONNX Runtime, 4-thread CPU). The Tier-1 model executes in 21.12 ms (47.4 FPS), exceeding the 45 FPS requirement. Tier-2 escalation runs in 67.61 ms (14.8 FPS). The Fisher-LDA projection requires only 9.8 kFLOPs (0.003 ms), adding negligible overhead. Since 88.4% of queries exit via Tier-1, the blended runtime is $0.884 \times 21.12 + 0.116 \times 67.61 \approx 26.5$ ms (>37 FPS).

Memory and Storage. The INT8 Tier-1 model occupies 22.05 MB in RAM and 15.13 MB for OTA updates. Packaged with ONNX Runtime and the Android UI, the standalone APK is 22.73 MB. The Tier-2 companion model occupies 85.3 MB. The total edge storage footprint is 108.03 MB, remaining within the 110 MB budget and preserving RAM for other drone/rover operations.

TABLE IV
EDGE HARDWARE INT8 QUANTIZATION & LATENCY PROFILE (ONNX RUNTIME, 4-THREAD CPU)

Edge Configuration	Params	INT8 Footprint	Lat. (ms)	Throughput
Tier 1: DINOv2 Small (Fast Path)	22.1M	22.05 (15.13†) MB	21.12	47.4 FPS
Tier 2: BioCLIP ViT-B (Deep Path)	86.6M	85.3 MB	67.61	14.8 FPS
Regularized Fisher Alignment	9.8K FLOPs	<0.01 MB	0.003	>10,000 FPS
BioDINO-Edge (Standalone APK)	**Native**	**22.73* MB**	**<22.00**	**>45.0 FPS**

†Lossless OTA payload. *Complete standalone Android APK; Tier 2 downloaded on-demand.

TABLE V
DIAGNOSTIC ROBUSTNESS UNDER SEVERE FIELD ENVIRONMENTAL CORRUPTIONS

Environmental Regime	Raw Acc.	Gate Rejection	Safe Acc.	Silent Error
Pristine (Clean Laboratory)	96.97%	0.00%	96.97%	3.03%
Solar Glare (Specular Highlight)	90.24%	0.00%	90.24%	9.76%
Canopy Shading (Harsh Shadow)	91.92%	0.67%	92.54%	7.41%
Motion Blur (Platform Shake)	78.45%	0.34%	78.72%	21.21%
Defocus Blur (Out of Focus)	49.16%	3.37%	50.17%	48.15%

Environmental Robustness. Table V evaluates stability under field corruptions. The system maintains 90.24% accuracy under solar glare and 91.92% under canopy shadows, demonstrating resilience to lighting gradients. Under severe defocus blur, accuracy drops to 49.16% because high-frequency lesion textures are attenuated. Deploying a lightweight pre-inference Laplacian variance filter (<0.5 ms) can reliably prompt operators to refocus before inference.

IV. DISCUSSION AND PRACTICAL DEPLOYMENT

Translating deep vision architectures from sanitized laboratory benchmarks into autonomous agronomic systems involves navigating the delicate tripartite trade-off between diagnostic sensitivity, inference latency, and computational autonomy:

Risk-Aware Autonomous Scouting and Decision Gating. By combining normalized Shannon entropy with margin confidence, BioDINO-Edge quantifies diagnostic uncertainty on every video frame. On autonomous agricultural UAVs, high-uncertainty frames trigger localized re-inspection or alert a human agronomist, preventing unwarranted chemical dispensing and optimizing agrochemical usage.

The Host-Context Loss Dilemma: Why Lesion Cropping Fails. While cropping bounding boxes tightly around lesions is often assumed to reduce background clutter, doing so across all 27 PlantDoc categories via DINOv2 attention rollout degraded 27-way accuracy from 58.73% to 57.14% (macro F1: 57.95% $\rightarrow$ 56.79%). In-depth error analysis reveals that tight cropping strips away macro-morphological phyllotaxy (e.g., compound vs. simple leaflet layout, venation symmetry), inducing severe cross-species confusion between homologous necrotic spots on different crops (e.g., Early Blight on tomato vs. potato). Preserving macroscopic host context is as vital for taxonomic transfer as isolating localized lesions.

Closed-Form Adaptation as an Antidote to Edge Backpropagation. While gradient-based meta-learning and fine-tuning pipelines require intensive GPU backpropagation, both Fisher alignment and transductive self-calibration in BioDINO-Edge are solved in closed form. Edge micro-controllers adapt to newly emerging pathogens in sub-second closed-form execution

without thermal throttling, memory fragmentation, or excessive computational overhead.

Deployment in Connectivity-Constrained Rural Environments. Intermittent rural connectivity is directly accommodated by our two-tier modularity. The 22.73 MB standalone APK provides immediate offline diagnostic utility upon initial field deployment, while the 85.3 MB companion downloads on demand when broadband connectivity becomes accessible.

V. Conclusion

We introduced **BioDINO-Edge**, an adaptive dual-foundation framework uniting self-supervised visual embeddings (DINOv2 Small) with organismal taxonomic vision-language supervision (BioCLIP ViT-B). By coupling high-dimensional Regularized Fisher-LDA with closed-form Soft-Anchored Transductive Self-Calibration and predictive Shannon uncertainty gating, it sets a new state of the art for few-shot plant pathology: **98.72% peak accuracy**, **98.41% peak F1-score**, and a 30-episode mean of **97.37% ± 0.73%** on PlantVillage, while restricting false-negative disease errors to **0.604%** and cross-crop confusion to **0.011%**. The entire system compiles into an ultra-lightweight 22.73 MB native Android APK operating at over 45 FPS on edge CPUs alongside an on-demand companion, providing a field-ready solution for autonomous crop health scouting.

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